MIL-I-45177A
1 March 1966
SUPERSEDING
MIL-I-45177
3 September 1958

MILITARY SPECIFICATION

INSTRUMENTS, TRACER, SURFACE ROUGHNESS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

* 1.1 Scope.- This specification covers the minimum essential requirements for instruments using stylus tracers with electrical amplification to measure surface roughness in the terms prescribed in ASA B46.1.

1.2 Classification.- Tracer type surface roughness instruments shall be of the following classes as specified (see 6.2).

Class 1: General purpose for shop use.
Class 2: Precision for laboratory use.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-585	Box, Wood, Wirebound
PPP-B-591	Box, Fiberboard, Wood-Cleated
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-T-76	Tape, Pressure Sensitive Adhesive Paper, Water Resistant, For Carton Sealing

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Military

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Standards Association (ASA)

B46.1 Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 General.- Instruments conforming to this specification shall have means for electrical amplification of the motion of a stylus held perpendicular to a surface over which it is traversed, and shall provide meter readings of surface roughness values which represent the arithmetical average of the deviations from a mean line in microinches as described in ASA B46.1.

3.2 Detail.

3.2.1 Stylus.

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* 3.2.1.1 Profile.- The profile of the stylus shall be either conical or pyramidal with a spherical tip. The included angle of the cone or pyramid shall be 90° with a tolerance of minus 5°.

3.2.1.2 Stylus tip radius.- The radius of the stylus tip shall be as follows:

Class 1 instruments:	500 ± 100 microinches
Class 2 instruments:	100 ± 50 microinches

3.2.1.3 Stylus force.- The maximum stylus force shall be as follows:

Class 1 instruments:	2.5 grams
Class 2 instruments:	0.3 gram

* 3.2.2 Tracer head.- The tracer head shall support the stylus in a manner which prevents lateral movement and shall be designed for both internal and external measurement of flat or radial surfaces with or without attachments.

* 3.2.3 Tracer head supports.- Instruments complying with this specification may employ the following means for establishing a reference line from which the actual deviations are measured:

(a) Use of skids or pivoted shoes which ride over the surface being tested. Skids shall be adaptable to flat and radial surfaces.

(b) Use of a precision slideway or guide which imparts a straight or appropriately curved motion to the stylus support.

* 3.2.4 Radius of skids.- If a single skid is employed to provide a reference surface, it shall have a radius in the direction of a trace of at least 50 times the roughness - width cutoff. If two operative skids transverse to the stylus are used, their radii shall be not less than 8 times the roughness - width cutoff.

* 3.2.5 Movement of tracer.- Unless otherwise specified, each instrument shall be provided with or incorporate a motor driven device for moving the tracer head at uniform speed. The device

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shall provide adjustment to vary the length of traverse and the movement may be reciprocating. The effect of reversals and motor vibration must be minimized to the extent that it does not influence the meter or recorder beyond the accuracy requirements of the instrument. The length of traverse shall be as follows:

Class 1 instruments:	Adjustable from 1/16 to 1-1/2 inches.
Class 2 instruments:	.07 min. for .01 roughness width cutoff.
	.15 for .03 roughness width cutoff.
	.30 for .10 roughness width cutoff.
	1/2 inch minimum for graphic recording.

3.2.6 Electrical characteristics.- Class 1 instruments may employ a self-generating type pickup (moving coil, vacuum tube transducer or crystal) in which continuous displacements of the stylus generate voltages which are amplified and fed to a meter. Class 2 instruments shall employ a carrier modulating pickup (differential transformer or condenser) type which is responsive to a static displacement of the stylus.

3.2.6.1 Overall response.- The instrument response shall include frequencies corresponding to surface irregularity widths from the designed long wavelength cutoff point (roughness width cutoff point) to as small as 0.0001 inch. The output for a sinusoidal, constant-amplitude, mechanical input shall be flat within 10 percent over the band pass region except in the immediate vicinity of the cutoff wavelength.

3.2.6.2 Long wavelength cutoff.- All class 1 instruments shall provide for a long wavelength cutoff point corresponding to a roughness width of 0.03 inch and when specified, shall be capable through adaptation to accommodate cutoff widths of 0.003 to 0.010 inches. Class 2 instruments shall provide for a long wavelength cutoff point corresponding to a roughness width of 0.01, 0.03 and 0.1 inches and when specified, shall be capable through adaptation to accommodate a cutoff width of 0.003 inch.

3.2.6.3 Attenuation rate beyond cutoff.

3.2.6.3.1 Pass band of system.- The pass band of the measuring system is defined as the wavelength (or frequency) range between the 3 decibel (db) down points of the system's wavelength (or frequency) response characteristic.

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3.2.6.3.2 Short wavelength attenuation rate.- The attenuation rate outside of the pass band on the short wavelength end may be of any value consistent with good design, except that attenuation shall be sufficient to prevent sensitivity of the instrument to external disturbances at high frequency. In no case may the amplifier characteristic rise above the condition of normal (midband) gain.

3.2.6.3.3 Long wavelength attenuation rate.- The minimum attenuation rate beyond the pass band on the long wavelength end shall be the equivalent of that produced by two R-C networks, with equal time constants, in series. This describes a system in which the straight part of the attenuation curve has a characteristic slope of 12 decibels per octave.

3.2.7 Indicating instrument.- Class 1 instruments shall be equipped with an indicating meter. Class 2 instruments shall be equipped with both an indicating meter and a graphic recorder.

3.2.7.1 Unit of measurement.- The indicating meter shall provide readings of the arithmetical average deviation from the mean line in microinches. Graphic recorders shall provide a representation of surface profile on co-ordinate paper which enables the reading of total deviation in microinches.

3.2.7.2 Characteristics.- The circuit or instrument used for obtaining average readings shall have the following characteristics:

(a) Response time.- The time to attain 95 percent of final deflection when driven by a suddenly applied signal shall be no shorter than 0.5 seconds or 10/f seconds, whichever is the longer period. The frequency (f) in cycles per second corresponds to the long wavelength cutoff point.

(b) Damping.- The coefficient of damping shall be not less than 0.6 critical.

(c) Scale linearity.- The scale of the indicating meter shall be linear from 20 percent of full scale to full scale. Departures from linearity below 20 percent of full scale shall be held to a minimum.

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(d) Integrated roughness meters.- For instruments having meters that indicate integrated roughness over a fixed length of trace, the requirements for response time and damping do not apply (see "a" and "b" above).

(e) Range of measurement.-

Class 1 instruments: 0 to 500 microinches.
Class 2 instruments: 0 to 200 microinches.

3.2.8 Instrument accuracy.- The values obtained from the instrument shall not deviate from the values of a mechanical input of known amplitude, within the amplitude and frequency range of the instrument by more than 10 percent on class 1 instruments and by more than 5 percent on class 2 instruments.

3.2.9 Precision reference specimens.- Each instrument will be furnished with precision reference specimens having an assigned roughness height and roughness width cutoff rating. Assigned value shall not vary from the actual measured value by more than 1 microinch or 3 percent whichever is the larger.

(a) Class 1 instruments shall be furnished with either of the following:

(1) Precision ground or etched glass specimens correlated with samples rated by class 2 instrument.

(2) Precision ruled metal specimens with a profile shown in Figure 1 calibrated by direct measurement or by a class 2 instrument.

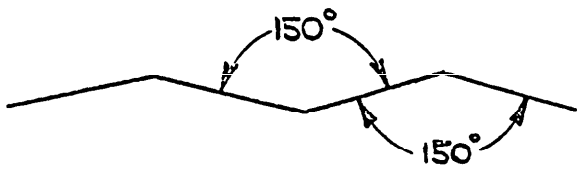


Figure 1. Profile of Precision Ruled

(b) Class 2 instruments may be checked with a pair of gage blocks. However, for a convenience check, a rated specimen calibrated by interferometric means shall be provided.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Acceptance tests.- Each instrument shall be subjected to the following tests or measurements as described under 4.3:

(a) Examination of product.
(b) Measurement of stylus.
(c) Performance.
(d) Measurement of reference specimens.
(e) Accuracy.

If one or more of the tests do not meet the requirements specified, it shall be cause for rejection.

4.3 Test methods.

4.3.1 Examination of product.- The instrument shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and general characteristics.

* 4.3.2 Measurement of stylus.- Profile and tip radius of the stylus shall be checked in an optical projector with a minimum power of 100X. The included angle of the stylus profile and stylus tip radius shall be within the limits prescribed in 3.2.1.1 and 3.2.1.2.

4.3.3 Performance.- The instrument shall be operated to test general performance, and repeatability of readings on a minimum of five samples of each of five different roughness height values representative of the entire required range of the instrument. For repeatability five readings will be obtained from each sample.

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4.3.4 Measurement of reference specimens.- Reference specimens furnished with the instrument in accordance with 3.2.9 shall be checked to determine compliance with the assigned value.

4.3.4.1 Precision ground or etched glass specimens.- Glass specimens shall be measured with a class 2 instrument which has been calibrated against other precision standards of known accuracy.

4.3.4.2 Precision ruled metal specimens.- Ruled metal specimens shall be correlated with other similar specimens which have been measured by "taper section" or other comparable methods. Correlation shall be accomplished using a class 2 instrument.

4.3.5 Accuracy.- To determine compliance with the specified accuracy tolerance in 3.2.8 five readings shall be taken from a minimum series of five calibrated samples ranging from 2 microinches to 250 microinches for class 1 instruments and from 1 microinch to 150 microinches for class 2 instruments. The total variation in readings shall not exceed the tolerance range.

* 4.3.6 Inspection of preparation for delivery.- Preservation, packaging, packing and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

* 5.1.1 Level A.- Instruments shall be preserved in accordance with MIL-P-116, method II, and packed in containers conforming to PPP-B-636. Unit quantity shall be one each.

5.1.2 Level C.- Instruments shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Intermediate packaging.- Unless otherwise specified by the procuring activity, the quantity of unit packs in the intermediate pack shall be at the option of the manufacturer.

5.3 Packing.

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* 5.3.1 Level A.- Instruments preserved and packaged as specified in 5.1.1 and 5.2 shall be packed in exterior-type shipping containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621 or PPP-B-636. As far as practical, exterior containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit, intermediate, or exterior container conforms to PPP-B-636 and is sealed at all joints and seams, including manufacturer's joint, with tape conforming to PPP-T-76.

* 5.3.2 Level B.- Instruments preserved and packaged as specified in 5.1.1 and 5.2 shall be packed in domestic-type exterior containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621 or PPP-B-636. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. When fiberboard containers are used, the fiberboard shall meet PPP-B-636.

5.3.3 Level C.- Packages which require over-packing for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery.

5.4 Marking.- Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Instruments covered by this specification are intended to be used for the determination of surface finish requirements on Government procured material. The use of class 2 instruments shall usually be limited to the inspection of tools, gages or other precision material when predominence of finishes are fine, critical or both.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class of instrument (see 1.2).
(c) If a motor-driven tracer is not required (see 3.2.5).
(d) When adapter is required for additional cutoff widths (see 3.2.6.2).

6.3 Identification of changes.- The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - WP
Air Force - 67

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Navy - SH, WP
Air Force - 67

User activities:
Army - None
Navy - MC
Air Force - None

Preparing activity:
Army - MU

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